

ALYSSA I. AGARD

(732) 234-9640 | aa@alyssaagard.dev

<https://alyssaagard.dev> | <https://www.linkedin.com/in/alyssaagard/>

RESEARCH PROFILE

Historian and defense policy researcher. Master of Public Policy candidate at Rutgers University, concentrating in political processes and institutions with a research focus on foreign affairs and defense policy, and Manager of Agard Research Associates Inc., a nonpartisan 501(c)(3) research institute in New Jersey. Present research attends to the problem of strategic stability in an age of intelligentized warfare: escalation dynamics under incomplete information, PLA counterspace doctrine and the vulnerability of space-based nuclear command and control, the logic of systems destruction warfare, and the organizational and cognitive habits of those who decide. Methods proceed by doctrinal genealogy, structured focused comparison, and process-tracing, joined to quantitative and geospatial modeling and to purpose-built simulation software. Where the established instruments of the field will not answer a question, I build my own. This work rests upon a formation in history, anthropology, and archaeology, with an abiding regard for material culture and the *longue durée*, for strategy is a human artifact and yields its logic most readily to those who study it as such.

EDUCATION

Rutgers University, Edward J. Bloustein School of Planning and Public Policy

Master of Public Policy

New Brunswick, NJ

Expected May 2027

- Concentration: Political Processes and Institutions
- Area of Research: Foreign Affairs and Defense Policy
- Selected graduate coursework: advanced statistics, quantitative methods, public finance and economics

Georgian Court University

Bachelor of Arts in History, Magna Cum Laude

Lakewood, NJ

August 2025

- Concentration: Contemporary American History and Foreign Relations
- Chapter President, Phi Alpha Theta National History Honor Society (2024 to 2025)
- Additional study in anthropology and archaeology, engaging material culture, social organization, and comparative approaches to institutional development

Brookdale Community College

Associate in Arts in Political Science and Humanities

Lincroft, NJ

2023

Additional and Self-Directed Study

- Theoretical and quantum physics: self-directed grounding in the physics governing orbital systems, undertaken in support of constellation degradation and counterspace modeling, with continuing study of quantum physics
- Financial and econometric modeling in code, applied to questions of political economy
- New Jersey and federal special education and general education coursework, including lesson planning and curriculum design

RESEARCH AND PROFESSIONAL EXPERIENCE

Agard Research Associates Inc.

Chairman and President of the Board of Directors; Chief Executive Officer

East Windsor, NJ

March 2025 to Present

- Established the organization and secured 501(c)(3) tax-exempt status with public charity designation for a nonpartisan interdisciplinary research institute producing scholarly research, policy analysis, and public programming.
- Direct the research agenda across defense policy and strategic stability, institutional design, and New Jersey employment and labor policy, holding outputs to the standards of academic peer discourse whilst keeping them legible to practitioners, policymakers, and the engaged public.

- Author and publish the institute's policy analyses, including reports on federal documentary proof of citizenship proposals, the New Jersey Pay and Benefits Transparency Act, and the state's temporary disability and family leave insurance programs.
- Apply quantitative and spatial methods in R, SPSS, SQL, Python, and ArcGIS to support rigorous and replicable analysis across research and policy products.
- Direct editorial, public relations, and media operations, including custom media design, publication delivery and output, and advertising campaign development.
- Design and deliver humanities-anchored K-12 educational programming and professional development workshops for New Jersey small businesses and private schools on cultural competency, social and emotional learning, and regulatory compliance under FERPA, HIPAA, OSHA, and EEOC standards.
- Direct programmatic operations spanning employment assistance prioritizing veterans and underserved communities, civic education, and behavioral health and trauma support partnerships.
- Manage organizational budgets, governance and compliance frameworks, IT infrastructure, and consulting engagements furnishing policy guidance to private and nonprofit clients.
- Maintain the organization's standing as a registered Microsoft Partner and Developer, under which its open-source Office add-ins are developed and released.

South China Sea NewsWire

Manager of Strategic Outreach and Partnerships

July 2026 to Present

- Direct strategic outreach and partnership development for an independent publication covering maritime security, regional strategy, and political affairs in the South China Sea.
- Serve as media outreach specialist, cultivating relationships with researchers, practitioners, and institutional partners, and coordinating editorial visibility across channels.
- Contribute published commentary on Chinese regional strategy and the diplomacy of maritime disputes.

Monmouth Battlefield State Park

Archival Research and Policy Intern

Manalapan, NJ

February 2026 to Present

- Conduct primary source and archival research on battlefield operations, military strategy, and Revolutionary War history in support of public programming and institutional interpretation.
- Synthesize archival findings into clear and usable outputs for public-facing interpretation and internal institutional needs.
- Analyze state and local policy frameworks governing historic site management, cultural preservation, and public lands stewardship.

PUBLICATIONS

Articles Under Peer Review

Agard, Alyssa I. "The Intelligentized Security Dilemma: Systems Destruction Warfare, Technological Entanglement, and the Erosion of Strategic Stability." 2026. Under peer review at Space and Defense. Preprint available.

Argues that the convergence of the PLA's Systems Destruction Warfare doctrine with artificial intelligence, quantum sensing, hypersonic weapons, and counterspace capabilities creates an intelligentized security dilemma characterized by compressed decision timelines, the entanglement of nuclear, conventional, and space architectures, and nonlinear escalation pathways that render Cold War stability models obsolete. Traces PLA doctrinal evolution from mechanization through informatization to intelligentization, evaluates the internal frictions constraining modernization, and proposes domain-specific governance mechanisms tailored to the compound dynamics identified.

Agard, Alyssa I. "Deterrence in Orbit: PLA Counterspace Doctrine, JADO, and the Vulnerability of Space-Based Nuclear Command and Control." 2026. Under peer review at Journal of Advanced Military Studies. Preprint available.

Examines whether Joint All-Domain Operations accounts for the nuclear escalation risks that PLA counterspace capabilities pose to dual-use space-based NC3 infrastructure. Introduces the space-nuclear firewall as an analytical concept describing JADO's implicit assumption that conventional space operations and nuclear command-and-control functions can be managed

within a unified domain framework without triggering escalation, argues that this assumed separation is operationally fictitious, and proposes a dedicated Space-Nuclear Integration Cell within the JADO planning architecture.

Policy Analyses and Institutional Reports

Agard, Alyssa I. "Documentary Proof of Citizenship and Voter Identification in Federal Elections: A Policy Analysis of the SAVE America Act, Related Executive Actions, Judicial Rulings, and Implications for New Jersey Voters." Agard Research Associates Inc., 2026.

Distinguishes what has been enacted, proposed, and blocked among federal proposals to require documentary proof of citizenship for voter registration and photo identification for voting, establishing that no new federal requirement is in force for the November 2026 election, and demonstrating how qualifying documents trace back to a birth record or passport, creating an uneven burden for citizens whose legal name has changed since birth.

Agard, Alyssa I. "Pay and Benefits Transparency in New Jersey Employment: A Policy Analysis of the New Jersey Pay and Benefits Transparency Act (P.L. 2024, c. 91), Its Proposed Implementing Regulations, the Federal Legal Landscape, and Measured Effects of Pay Transparency Policy." Agard Research Associates Inc., 2026.

Separates the statute now in force from the proposed departmental rules that remain non-binding, explains coverage and penalty assessment, examines what the state's first enforcement initiative revealed about early compliance among New Jersey's largest employers, and situates the law within the wider pay equity framework from the federal Equal Pay Act and Title VII to the Diane B. Allen Equal Pay Act.

Agard, Alyssa I. "Temporary Disability and Family Leave Insurance in New Jersey: A Policy Analysis of New Jersey's Wage-Replacement Programs, the 2026 Benefit Parameters, the Job-Protection Amendments of P.L. 2025, Chapter 279, and the Federal Framework." Agard Research Associates Inc., 2026.

Sets out the 2026 parameters now in force for the two state-administered wage-replacement programs, centers on the amendments attaching reinstatement rights to TDI and FLI leave and expanding the New Jersey Family Leave Act, corrects the record on two widely misstated points concerning the benefit maximum and the enacted employer coverage threshold, and places the state programs within the federal framework.

Agard, Alyssa I. "Decision-Making in Defense Policy: Institutions, Bias, and Strategic Accountability." Agard Research Associates Inc., 2025.

Examines how institutional design, cognitive bias, and accountability structures shape defense policy outcomes, drawing on organizational theory and case evidence from United States national security decision-making.

Commentary and Public Writing

Agard, Alyssa I. "What the Beijing Summit Did Not Say About the South China Sea." South China Sea NewsWire, 2026.

<https://southchinaseanewswire.com/what-the-beijing-summit-did-not-say-about-the-south-china-sea/>

Manuscripts in Progress

Agard, Alyssa I. "Coordination Tax: Doctrinal Latency, the Limits of Sensor-Shooter Compression, and Fires Clearance in Contested Battlespace." Manuscript in development, intended for academic press submission.

Advances the concept of the coordination tax, defined as doctrinal and procedural delay produced by Fire Support Coordination Measure clearance independent of network speed, and argues that at the tactical artillery and counter-battery level, conditional on adequate magazine depth and outside saturation electronic warfare environments, this tax is the dominant first-order constraint on fires effectiveness. Draws on the doctrinal genealogy of the FSCM framework refined during the AirLand Battle era, structured comparison of paired Ukrainian and Israeli engagement cases, and process-tracing of open-source timing anchors, and proposes a doctrinal redesign grounded in delegated clearance authority.

APPLIED RESEARCH PROJECTS AND ANALYTICAL SOFTWARE

Formal simulations delivered as interactive software, structured rather than fitted, each validated against documented empirical cases with sources auditable within the application. Code, documentation, and methodology are public and reproducible.

The Primacy Premium: Conditional Forecasts of Defense and Commercial Markets Under Chinese Military

Primacy

2026

Python modeling pipeline; in development

A conditional forecasting pipeline, calibrated on seven decades of IMF and SIPRI data, modeling how global defense spending, arms transfers, and commercial risk pricing would reallocate under three distinct pathways to Chinese military primacy by 2035. Reports full probability distributions across thousands of simulations per pathway, parses and validates official datasets, estimates response parameters by HAC regression, and exports analysis-ready tables for Power BI and Tableau.

<https://alyssaagard.github.io/primacy-premium-preview/>

The Coordination Tax: Field Artillery Engagement Simulation

2026

Python, Streamlit, and static client-side build

Models field artillery engagement effectiveness across four reference platforms (M777A2, M109A7 Paladin, M142 HIM-ARS, and 2S33 Msta-SM2) through a dimensionless framework composing Amdahl's Law with exponential target perishability decay. Features platform-specific weapon behavior, twenty-two academic citations, seven empirical scenario presets drawn from RUSI commentary on the Russo-Ukrainian War, and live-updating visualizations including a tactical engagement diorama, phase-decomposition timeline, and perishability decay curve. Companion analytical instrument to the working paper of the same name.

<https://alyssaagard.github.io/coordination-tax/>

ATLAS: Advanced Threat-Level Analysis and Simulation

2026

Browser-based crisis wargaming platform, React

A research instrument for testing escalation thresholds, signaling dynamics, and the structural sources of inadvertent escalation under incomplete information, modeled on the NSC simulation methodology used at the National Defense University, Naval War College, and RAND. Users navigate multi-turn crisis scenarios through the DIME framework while an algorithmic White Cell adjudicates outcomes, a Red Cell generates adversary responses, and an eight-level escalation ladder tracks the trajectory toward or away from conflict. Ships with Taiwan Strait and NATO Baltic flank scenarios, a custom scenario builder, randomized injects, force disposition tracking, and a comprehensive After Action Review. Fully standalone with no external dependencies.

<https://alyssaagard.github.io/atlas-wargame/>

Degraded Constellation: Counterspace Geospatial Simulation

2026

Interactive geospatial simulation

Models the effects of PLA antisatellite operations on United States missile warning coverage during a Taiwan Strait contingency. Users strike individual SBIRS and OPIR satellites to observe how constellation degradation simultaneously erodes conventional theater awareness and nuclear early warning, generating the escalation ambiguity that Acton's damage-limitation framework identifies as most dangerous. Companion analytical tool to "Deterrence in Orbit," with parameter sets supporting scenario variation across attack timing, target selection, and constellation resilience assumptions.

<https://agard-research-associates.github.io/degraded-constellation/>

PRODUCTION SOFTWARE

Developed and released through Agard Research Associates Inc. as a registered Microsoft Partner and Developer.

Layout Toolkit (V1, V2, and Google Docs edition)

2026

Microsoft Word add-in, Office.js API

Built in JavaScript, HTML, and CSS on the Office.js API, bundled with Webpack and deployed as a static site. Provides six document templates, saveable brand profiles, page layout controls including columns and margins, and visual folding guides for print production. A single codebase targets Word for desktop, Mac, and the web, using runtime guards to absorb platform-specific API differences. Released free and open-source under the MIT License.

Cachet

In development

Cross-platform email signature management add-in for Microsoft Outlook and Gmail

Gives administrators centralized control over branded signatures across an organization through three authoring modes: a field-based form for rapid setup, a visual editor modeled on Outlook's native signature interface, and a raw HTML editor for

full design control. Supports configurable name, title, contact, and social fields, uploaded logos with adjustable sizing, written or linked legal disclaimers, and drag-and-drop layout ordering, with automatic assignment to users.

INVITED PRESENTATIONS AND MEDIA

Guest, NucleCast, the podcast of the ANWA Deterrence Center, hosted by Dr. Adam Lowther. Topic: strategic stability and Chinese military modernization.

Editorial and media production, Agard Research Associates Inc., 2025 to Present. Direction of public relations and publication output, including custom media design, delivery, and advertising campaign development.

WORKSHOPS AND CIVIC EDUCATION

"Ballot, Power, and Democracy in New Jersey: A Community Workshop on New Jersey Polling and Civic Participation." Agard Research Associates Inc., 2026.

A nonpartisan civic education workshop examining how New Jersey's county line ballot system shaped state politics for decades, drawing on peer-reviewed research by Prof. Julia Sass Rubin of Rutgers University. Three presentation tracks serve general audiences, students, and English language learners.

<https://agardresearchassociates.org/ballot-power-and-democracy-in-nj>

Professional development workshops for New Jersey small businesses and private schools on cultural competency, social and emotional learning, and regulatory compliance under FERPA, HIPAA, OSHA, and EEOC standards. Agard Research Associates Inc., 2025 to Present.

RESEARCH METHODS

- Doctrinal genealogy and comparative analysis of United States and foreign military doctrine, including FM, ATP, and JP publications and PLA source materials
- Structured focused comparison and Mill's method of difference
- Process-tracing using open-source operational and timing evidence
- Archival and primary source research
- Constructive simulation and wargame design for escalation and decision-making research
- Quantitative analysis: descriptive and inferential statistics, HAC regression, Monte Carlo simulation, sensitivity analysis
- Geospatial modeling and simulation
- Financial and econometric modeling in code, applied to questions of political economy
- Applied software development supporting reproducible defense and policy research

TECHNICAL SKILLS

Statistical and Quantitative: R (advanced), SPSS (advanced), Python (numpy, matplotlib), SQL; data cleaning and manipulation, descriptive and inferential analysis, simulation and numerical modeling.

Geospatial: ArcGIS; geospatial modeling and cartographic analysis.

Development: JavaScript, React, HTML and CSS, Python, C++, Office.js API, Webpack, Git and GitHub; static site deployment; cybersecurity fundamentals; IT infrastructure administration.

Platforms and Tools: Microsoft 365 Enterprise Suite, Azure, PowerShell, Power BI, Tableau, Streamlit, Microsoft Excel (advanced), Google Workspace.

Design and Media: Web and media design, editorial production, publication layout and delivery, advertising campaign development.

Research and Communication: Archival research, policy analysis, technical writing, data visualization, literature synthesis, survey design, report development.

LANGUAGES

Latin: Fluent reading proficiency.

Russian, Mongolian, and Mandarin Chinese: Self-directed study for primary source research; limited working proficiency.

HONORS AND AWARDS

- Magna Cum Laude, Georgian Court University (2025)
- Chapter President, Phi Alpha Theta National History Honor Society, Georgian Court University (2024 to 2025)

CERTIFICATIONS AND TRAINING

- CITI Program (Collaborative Institutional Training Initiative): Social and Behavioral Responsible Conduct of Research, Human Research (SBE), Research Security Training, Research Study Design, Conflicts of Interest, and additional modules
- Registered Behavior Technician (RBT), Behavior Analyst Certification Board
- New Jersey Substitute Teacher Certification
- Microsoft Partner and Developer, registered through Agard Research Associates Inc.

PROFESSIONAL AFFILIATIONS

- American Historical Association (AHA), Member and Donor (2024 to Present)
- Monmouth County Historical Association (MCHA), Member and Donor (2023 to Present)
- Phi Alpha Theta, National History Honor Society, Member (2024 to Present)
- New Jersey Association of Professional Mediators (NJAPM), Member (2025 to Present)
- National Association for the Advancement of Colored People (NAACP), Member

RESEARCH IDENTIFIERS AND REPOSITORIES

ORCID: <https://orcid.org/0009-0008-5783-2437>

Code and data repositories: <https://github.com/alyssaagard>

Research: <https://research.alyssaagard.dev>

Projects: <https://projects.alyssaagard.dev>

ADDITIONAL PROFESSIONAL EXPERIENCE

Collier Youth Services and New Jersey Public School Districts

New Jersey

Certified Substitute Teacher, Behavioral Special Education

September 2023 to Present

- Deliver individualized instruction and behavioral support to students with disabilities in therapeutic and public school settings, adapting content to IEP and 504 Plan goals across general and special education classrooms.
- Manage classroom behavior through positive behavior supports, consistent routines, and de-escalation strategies.

Sunny Days Sunshine Center

Manalapan, NJ

Registered Behavior Therapist, Early Intervention

June 2024 to January 2026

- Implemented individualized treatment plans within Applied Behavior Analysis frameworks for infants, toddlers, and young children with autism spectrum disorder and developmental delays.
- Collected, graphed, and analyzed behavioral data each session in collaboration with Board Certified Behavior Analysts to adjust interventions against mastery targets.